# xjyutping: Jyutping above Traditional Chinese characters

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**Abstract**

This package adds Cantonese Jyutping (粵拼), the romanisation of Cantonese devised by the Linguistic Society of Hong Kong (LSHK), above Traditional Chinese characters. It is aimed at anyone preparing Cantonese material in LaTeX, such as teaching notes, lyrics or poetry, where the pronunciation of each character should be shown above it. The reading of each character is chosen from the surrounding words, so 行 is read *hong4* in 銀行 but *haang4* in 行路. Every character sits in a cell of the same width, which is wide enough for its Jyutping, so the text stays evenly spaced and no Jyutping touches its neighbours or the line above. The package works with XeLaTeX through xeCJK and with LuaLaTeX through LuaTeX-ja, including the ctex package and classes under either engine.

## 1 Getting started

To annotate a document,

1. Load ctex, xeCJK or luatexja-fontspec and choose a Traditional Chinese font
2. Load xjyutping
3. Put the text inside a `jyutpingscope` environment
4. Compile with `xelatex` or `lualatex`

A minimal document is,

```
\documentclass{article}
\usepackage[fontset=none]{ctex}    % for xelatex or lualatex
\setCJKmainfont{Songti TC}
\usepackage{xjyutping}
\begin{document}
\begin{jyutpingscope}
我哋去銀行，行路返屋企。校長喺學校長大。
\end{jyutpingscope}
\end{document}
```

ngo5 dei6 heoi3 ngan4 hong4 haang4 lou6 faan1 uk1 kei2 haau6 zoeng2 hai2 hok6 haau6 zoeng2 daai6
我 哋 去 銀 行，行 路 返 屋 企。校 長 喺 學 校 長 大。

The ctex package sets up Chinese typesetting for either engine. Instead, xeCJK can be loaded under XeLaTeX, or luatexja-fontspec (with `\setmainjfont`) under LuaLaTeX. If none of them is loaded, xjyutping loads xeCJK or LuaTeX-ja itself. Any Traditional Chinese font can be used, and the Jyutping is set in the main Latin font of the document unless another font is chosen (Section 4).

# 2 Commands

The commands are mainly divided into two groups, the commands that annotate text (Section 2.1) and the commands that choose a reading (Section 2.2).

## 2.1 Annotating text

**\begin{jyutpingscope}[⟨*options*⟩] ... \end{jyutpingscope}** annotates every Chinese character in the block. The block ends the paragraph, and its line spacing is raised where needed so that the Jyutping of one line never comes close to the line above.

**\xjyutping*[⟨*options*⟩]{⟨*text*⟩}** does the same for a piece of running text inside an ordinary paragraph.

For example,

```
香港人講\xjyutping*{廣東話}，寫\xjyutping*{繁體字}。
```

gwong2 dung1 waa2 faan4 tai2 zi6
香港人講 廣 東 話，寫 繁 體 字。

## 2.2 Choosing a reading yourself

Cantonese pronunciation changes with meaning and with colloquial tone change, and no word list is perfect. Hence there are three ways to set a reading by hand,

**\xjyutping[⟨*options*⟩]{⟨*characters*⟩}{⟨*readings*⟩}** gives the characters an explicit reading at that point only. It works inside a scope and on its own, and it also marks a word boundary.

**\setjyutping{⟨*character*⟩}{⟨*reading*⟩}** changes the reading a character takes when it is not part of a known word.

**\setjyutping{⟨*word*⟩}{⟨*readings*⟩}** adds a word or overrides the reading of a known one. Words always take priority over the defaults of single characters.

Readings are Jyutping syllables separated by spaces, with one syllable per Chinese character. Punctuation, Latin letters and commands among the characters take no syllable, so `\xjyutping{A行}{hong4}` reads 行 as *hong4*. Since `\setjyutping` is global and takes effect from where it appears, it can go in the preamble or in the middle of a scope.
In the following example, 當 means 'treat as', so it is read *dong3* rather than with its default *dong1* ('when'), while 家行 is a name read *gaa1 hang4*,

```
\setjyutping{當我}{dong3 ngo5}
\begin{jyutpingscope}
佢當我細路，\xjyutping{家行}{gaa1 hang4} 都係。
\end{jyutpingscope}
```

keoi5 dong3 ngo5 sai3 lou6 gaa1 hang4 dou1 hai6
佢 當 我 細 路，家 行 都 係。

### 2.3 Formatting inside words

Braces and formatting commands do not split a word, so a character can be highlighted without losing its reading in context. For example,

```
\begin{jyutpingscope}
佢喺銀\textbf{行} 返工，係校\textcolor{red}{長} 嘅朋友。
\end{jyutpingscope}
```

keoi5 hai2 ngan4 hong4 faan1 gung1 hai6 haau6 zoeng2 ge3 pang4 jau5
佢 喺 銀 行 返 工， 係 校 長 嘅 朋 友。

### 2.4 Switching off

The `\disablejyutping` command stops annotation until `\enablejyutping` or the end of the current group or environment. Use it for passages inside a scope that should stay plain.

## 3 How readings are chosen

The reading of each character is chosen in five steps,

1. First, the text is split into runs of Chinese characters. Punctuation, Latin text and most commands end a run, while spaces and line breaks between two characters do not, and neither do braces or formatting commands (`\textbf`, `\emph`, `\color`, `\textcolor`, size commands …). The text of a `\footnote` is a run of its own, and the text around the footnote mark carries on.

2. Then each run is divided into words from a list of about 104 000 Cantonese words. The division with the fewest words is chosen, then the one with the fewest single characters, then the one of the more common words, as counted in the word-frequency list of rime-cantonese and in the transcripts of WenetSpeech-Yue (步行街 is 步行 + 街, not 步 + 行街). Only a full tie goes to the longer final word.

3. A word takes its reading from the list, and the words set with `\setjyutping` are checked first and win a tie of step 2.

4. A character that is not part of a word takes the reading set with `\setjyutping` if there is one. Otherwise, some characters take their reading from the one or two characters after them, a few are read differently at the end of a run, and the rest take their default reading. For example, 呢 is the demonstrative *ni1* before a classifier (呢張床) but the particle *ne1* in 你呢？and 佢呢就走, while 咁 is *gam2* “like this” at the end of a run (就係咁) but *gam3* “so” before an adjective (咁大).

5. Lastly, a reading given with `\xjyutping{`⟨*characters*⟩`}{`⟨*readings*⟩`}` always takes priority, and it ends the run.

Traditional characters have several accepted shapes (為 and 爲, 裡 and 裏, 說 and 説, 衛 and 衞, 線 and 綫, 溫 and 温…). For word lookup these shapes are folded together, so a word is found whichever shape is typed.

### 3.1 Proofreading

About 4 400 characters have more than one common reading. When such a character is not settled by a word or by a setting of your own, its reading is only a guess. The `multiple` option formats these guesses, while the `debug` option writes every run to the log together with the other readings. For example,

```
\xjyutpingsetup{multiple=\color{red}, debug}
\begin{jyutpingscope}
佢當佢細路。
\end{jyutpingscope}
```

keoi5 dong1 keoi5 sai3 lou6
佢 當 佢 細 路。

The log then contains a line such as,

```
xjyutping> 佢 keoi5:s | 當 dong1:m(dong3) | 佢 keoi5:s | 細 sai3:w 路 lou6:w |
```

Each character is followed by its reading and a type, which is `w` for a reading from the word list, `u` for a reading you set, `m` for a guessed polyphone (with the other readings in brackets) and `s` for a character with a single common reading. Looking at this line, 當 means 'treat as' here, so `\setjyutping{當佢}{dong3 keoi5}` fixes it, while `\setjyutping{當}{dong3}` would make *dong3* the default for 當 on its own.

# 4 Options

Options can be given to `\usepackage`, to `\xjyutpingsetup` or in the optional argument of `jyutpingscope` and `\xjyutping`. The options are,

| Key | Default | Meaning |
|---|---|---|
| `ratio` | `0.45` | size of the Jyutping relative to the text |
| `vsep` | `1.05em` | height of the Jyutping baseline above the character's |
| `hsep` | `0.15em plus 0.4em` | space between cells, which is the least gap between two Jyutping plus stretch for justified lines |
| `width` | `auto` | cell width, which is `auto`, `natural` or a length |
| `font` | `\normalfont` | font of the Jyutping (font commands only) |
| `format` | | extra formatting, such as `\color{gray}` |
| `multiple` | | formatting for guessed polyphones |
| `fancy` | `false` | tones as pitch strokes (Section 5) |
| `linebreak` | `false` | keep the lines of the source (Section 6) |
| `align` | `justify` | alignment of the lines of a scope, which is `justify`, `left`, `centre` (or `center`) or `right`, also given alone, as in `[linebreak,centre]` |
| `debug` | `false` | log every run |

## 4.1 Cell width

With `width=auto`, every character in a block gets the same cell, which is wide enough for the longest Jyutping in that block. A length gives a fixed grid across the whole document, where a Jyutping longer than the cell widens its own cell rather than overlapping. On the other hand, `width=natural` makes each cell only as wide as its own character or Jyutping, which is tighter but uneven. For example,

```
\xjyutping*[width=auto]{香港中文大學}
\xjyutping*[width=natural]{香港中文大學}
\xjyutping*[width=2em]{香港中文大學}
```

hoeng1 gong2 zung1 man4 daai6 hok6
香 港 中 文 大 學

hoeng1 gong2 zung1 man4 daai6 hok6
香 港 中 文 大 學
hoeng1 gong2 zung1 man4 daai6 hok6
香 港 中 文 大 學

### 4.2 Size and font

The size, font and colour of the Jyutping are set with `ratio`, `font` and `format`. For example,

```
\xjyutping*[ratio=0.6, font=\sffamily, format=\color{gray}]{早晨，食咗飯未？}
```

zou2 san4 sik6 zo2 faan6 mei6
早 晨，食 咗 飯 未？

## 5 Fancy tones

With the `fancy` option (`\usepackage[fancy]{xjyutping}`), each tone number is shown in the same way as in Visual Jyutping. Specifically, it becomes a small stroke that traces the pitch of the tone, followed by a smaller tone number, which is raised for the two high tones and lowered for the others. The strokes follow the symbols of Visual Jyutping, where tone 1 is high, tone 2 rises to high, tone 3 is mid, tone 4 falls to low, tone 5 rises from low and tone 6 is low. For example,

```
{\Large\xjyutping*[fancy]{詩史試時市事}}
\begin{jyutpingscope}[fancy]
我哋去銀行，行路返屋企。校長喺學校長大。
\end{jyutpingscope}
```

si$^{-1}$ si$^{\nearrow 2}$ si$_{-3}$ si$_{\searrow 4}$ si$_{\nearrow 5}$ si$_{-6}$
詩史試時市事
ngo$_{\nearrow 5}$ dei$_{-6}$ heoi$_{-3}$ ngan$_{\searrow 4}$ hong$_{\searrow 4}$ haang$_{\searrow 4}$ lou$_{-6}$ faan$^{-1}$ uk$^{-1}$ kei$^{\nearrow 2}$ haau$_{-6}$ zoeng$^{\nearrow 2}$ hai$^{\nearrow 2}$ hok$_{-6}$ haau$_{-6}$ zoeng$^{\nearrow 2}$ daai$_{-6}$
我 哋 去 銀 行，行 路 返 屋 企。校 長 喺 學 校 長 大。

Since the strokes are drawn rather than taken from a font, they work with any `font` and take the colour of `format` and `multiple`. The spacing is also adjusted to the new shape. Specifically, the cells are measured with the strokes and small numbers in place, so `width=auto` cells come out a little wider, and the line spacing makes room for the raised numbers. Further, if the lowered numbers reach further down than the letters of the chosen font, the Jyutping is lifted by the difference so that it keeps its distance from the character. As with every option, `fancy` can be switched on or off for one scope, as in `\begin{jyutpingscope}[fancy=false]`. A syllable given without a tone number, such as `\xjyutping{唔}{m}`, is printed as it is.

## 6 Verse and lyrics

The `linebreak` option keeps the lines of the source, which suits verse and lyrics. Specifically, each line end inside `jyutpingscope` becomes a line break, as if it were written `\\`, while each blank line leaves an empty line and starts a new paragraph. The new paragraph follows the usual rules of the document, so it is indented by default and not indented under `\usepackage[parfill]{parskip}`, whose paragraph space is added to the empty line. The `align` option sets the lines flush left, centred or flush right, and a `\centering` inside the scope also works. For example,

```
\begin{jyutpingscope}[linebreak,centre]
床前明月光，
疑是地上霜。

舉頭望明月，
低頭思故鄉。
\end{jyutpingscope}
```

cong4 cin4 ming4 jyut6 gwong1
床 前 明 月 光，
ji4 si6 dei6 soeng6 soeng1
疑 是 地 上 霜。

geoi2 tau4 mong6 ming4 jyut6
舉 頭 望 明 月，
dai1 tau4 si1 gu3 hoeng1
低 頭 思 故 鄉。

An aligned scope starts a paragraph of its own. Line ends at the start and the end of the scope, after a `\\` and before `\begin`, `\end` or a new paragraph add no break, and a line ending in `%` joins the next one as usual. Since a line break also ends a run of characters, no word is looked up across two lines. However, the option only works in the environment, and only where the line ends are still in the source when the scope begins. Hence it has no effect in `\xjyutping*`, in a scope inside the argument of a command such as `\parbox{...}` or in a scope inside another scope without the option, since their text has already been read. Inside an environment such as `minipage`, on the other hand, it works.

# 7 XeLaTeX and LuaLaTeX

The readings, the options and the layout are the same under both engines, while the machinery differs. Under XeLaTeX, xeCJK hands each Chinese character to a hook in which xjyutping builds the cell of the character. Under LuaLaTeX, the rubies are built as the text is read, and a Lua function in `xjyutping.lua`, which must be installed next to `xjyutping.sty`, puts each cell together after LuaTeX-ja has laid out the line. This gives four differences under LuaLaTeX,

- the rules of LuaTeX-ja for punctuation widths and line breaks apply, so for example a line never breaks just before a dash (——) or an ellipsis (……),
- a paragraph can be of any length, and a paragraph of 40 000 characters compiles,
- compiling takes about 1.6 times as long as with XeLaTeX and
- text that comes from a macro is annotated with the options and the size of the place where it stands, but with the `\setjyutping` readings in force when its paragraph ends.

# 8 Troubleshooting

## 8.1 `\verb` inside a scope gives an error

The body of `jyutpingscope` and the argument of `\xjyutping*` are read in full before anything is typeset, in the same way as any argument. Hence `\verb` and verbatim environments cannot go inside, and in a `\url` or `\href` inside a scope a `%` must be written `\%`. Put verbatim material outside the scope.

### 8.2 Text is annotated one character at a time

Text that comes from a macro, an `\included` file or `\maketitle` is annotated one character at a time, without word context, and the debug log marks it `[no context]`. This is due to the package only seeing the name of a macro when it reads a scope. The same applies to a file read with `\input{`⟨*file*⟩`}` inside a scope when the file uses `\endinput`, `\verb`, verbatim environments, `\makeatletter` or catcode changes, since such a file is input normally instead of being read as part of the scope. Put `\xjyutping*` inside the macro, or a scope inside the included file.

### 8.3 A command's argument is annotated or left plain

The arguments of `\label`, `\ref`, `\cite`, `\index` and `\url`, the first argument of `\href`, the optional argument of `\hyperref`, `\hyperlink`, `\pdfbookmark`, `\includegraphics`, the cleveref commands and environment names are passed through untouched. Other commands need their Chinese argument in braces, as in `\textbf{行}` rather than `\textbf 行`.

### 8.4 The table of contents has no Jyutping

Section titles, captions and footnotes inside a scope are annotated, while the table of contents is only annotated if `\tableofcontents` is itself inside a scope, although an explicit `\xjyutping` in a title is annotated there as well. Running heads and PDF bookmarks are always plain.

### 8.5 A beamer frame title has no Jyutping

In beamer, a `\frametitle` inside a scope is typeset after the scope has ended, so it stays plain. Write `\frametitle{\xjyutping*{...}}` instead.

### 8.6 Underlines break the spacing

The underline and emphasis-mark commands of xeCJKfntef (XeLaTeX only) do not keep the cell spacing inside a scope. Use `\underline` or `\uline` from ulem instead.

### 8.7 A long paragraph exceeds the memory of TeX

A scope can hold well over 100 000 characters. However, under XeLaTeX one paragraph is limited by the memory of TeX to about 15 000 characters, or about 13 000 with `fancy`. Split the text into shorter paragraphs, or compile with LuaLaTeX, which has no such limit.

### 8.8 A `\setjyutping` in unused code still applies

A `\setjyutping` inside a scope is noticed while the text is read, so one inside `\iffalse`…`\fi` or an unused macro definition still applies to the text after it. Keep conditional settings outside the scope.

### 8.9 A character has no Jyutping

Characters missing from the data are typeset in their cell without Jyutping. Give such a character a reading with `\setjyutping` or `\xjyutping`, which both work for characters outside the data.

### 8.10 pdfLaTeX stops with an error

Only XeLaTeX and LuaLaTeX are supported, while pdfLaTeX is not. Compile with `xelatex` or `lualatex` instead.

# 9 Data and licences

The readings come from five groups of sources, which `tools/fetch-sources.sh` downloads at pinned commits,

- the *Cantonese Pronunciation List of Characters for Computers* (粵拼表) of the LSHK and the rime-cantonese dictionaries (both CC BY 4.0), which give the character readings, the defaults and most of the words, with rime-cantonese treated as authoritative,
- the word list of ToJyutping (BSD 2-Clause), which chooses between the readings that rime-cantonese gives a word, following Hong Kong usage (公園 *gung1 jyun2*), and adds 512 words,
- CC-Canto and the Cantonese readings of CC-CEDICT (both CC BY-SA 3.0, by Pleco), as distributed with Jyut Dictionary, which add 2 531 words, each only where it changes a reading and passes checks against rime-cantonese,
- the book data of 粵音資料集叢, which gives the readings of 640 rare characters that the others lack and
- the variant tables of OpenCC (Apache-2.0).

The word frequencies come from the list of rime-cantonese and from a count of how often each word is used in the 6.8 million transcribed utterances of WenetSpeech-Yue, which is kept in `tools/wenetspeech-yue-counts.tsv`. The tables at the top of `tools/build-data.py` hold the hand-checked corrections, and the script regenerates `xjyutping-chars.def` and `xjyutping-words.def` from these sources.

We measured the accuracy on seven corpora. On the Hong Kong Cantonese Corpus (HKCanCor), whose 161 045 characters of conversation were annotated with Jyutping by hand, the package reads 95.6% of the characters correctly on the half of the files kept out of the tuning, and 97.9% of the characters outside sentence-final particles and interjections. On the particles of CantoMap, which were transcribed by ear, it reads 97.5%, and on fresh sentences of SpiCE, MagicHub and WenetSpeech-Yue, where the systems disagree, its reading was judged right in 95.3% of the cases.

The package code (`xjyutping.sty` and `xjyutping.lua`) is released under the LaTeX Project Public License (LPPL) 1.3c. The data files are released under CC BY-SA 4.0, since they adapt the CC BY-SA word lists. The word list of ToJyutping is used under the BSD 2-Clause License, whose notice is reproduced in `LICENSE`. The readings taken from 粵音資料集叢 come from data published without a licence statement and are used with attribution. The word counts of `tools/wenetspeech-yue-counts.tsv` are counted from the transcripts of WenetSpeech-Yue (CC BY-NC 4.0), and no text of any corpus is included.

# 10 Acknowledgements

This package was inspired by the authors of the LaTeX package xpinyin by Qing Lee (李清), which puts Hanyu Pinyin above Simplified Chinese characters, and it follows the way xpinyin annotates characters through the `\CJKsymbol` hook of xeCJK. The `fancy` option was inspired by Visual Jyutping by Vincent Tam, whose tone symbols it draws, and by the Visual Cantonese Fonts (粵語字體) by Jon Chui / A3I Ltd. (canto.hk, docs.visual-fonts.com).

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- Aaron Tan for Jyut Dictionary,
- Pleco for CC-Canto and the Cantonese readings of CC-CEDICT, together with MDBG and the contributors of CC-CEDICT,
- Carbo Kuo (BYVoid) and the contributors of OpenCC,
- Kang Kwong Luke for the Hong Kong Cantonese Corpus (CC BY 4.0), as distributed with PyCantonese,
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- Beijing Magic Data Technology for the Guangzhou Cantonese Conversational Speech Corpus of MagicHub and
- the contributors of the Cantonese Wikipedia, whose corpora we used to tune the package and to measure its accuracy.